Student Assignment: free-space optics with turbulence and pointing errors

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1 Task 1 — The model

1.1 The three factors of Equation (2)

The received photocurrent is modelled as:

$$y = \eta h x + n$$

where η is the optical-to-electrical conversion coefficient, x is the transmitted signal, and n is additive Gaussian noise. The composite channel gain is expressed as:

$$h = h_a h_p h_l \tag{1}$$

The three components are:

1. Atmospheric path loss (h_l) Atmospheric path loss represents the attenuation caused by absorption and scattering from atmospheric molecules and aerosols. It follows the Beer–Lambert law:

$$h_l = e^{-\sigma(\lambda)L}$$

The attenuation coefficient and propagation distance are considered constant over the analysis window, making h_l deterministic. Using the parameters in Table I, the total attenuation over 3 km is $0.2208 \times 3 = 0.6624$ dB. Therefore:

$$h_l = 10^{-0.6624/10} \approx 0.8585$$

2. Geometric spread and pointing errors (h_p) As the optical beam propagates, it expands and spreads over the receiver plane. The beam radius is approximately:

$$w_z \approx \theta_{1/2} L = 0.66 \times 10^{-3} \times 3000 = 1.98 \text{ m}$$

Since the receiver aperture radius is only $a = 0.05$ m, a small fraction of the transmitted optical power is collected, even under perfect alignment. The maximum collected power fraction is:

$$A_0 = \text{erf}(v)^2$$

where:

$$v = \frac{\sqrt{\pi} a}{\sqrt{2} w_z}$$

Using the given parameters:

$$A_0 \approx 1.27 \times 10^{-3}$$

Random beam-centre displacement due to pointing errors further reduces the collected power. Therefore, h_p is modelled as a random variable.

3. Atmospheric turbulence (h_a) Atmospheric turbulence arises from refractive-index fluctuations caused by variations in atmospheric conditions. These fluctuations produce scintillation, resulting in random variations in the received optical irradiance. Consequently, h_a is modelled as a random variable using a log-normal distribution for weak turbulence. It is normalized such that:

$$E[h_a] = 1$$

Which factors are random, and why

Factor	Nature	Reason
h_l	deterministic	Set by the attenuation coefficient $\sigma(\lambda)$ and the fixed link length L. Weather, and hence $\sigma(\lambda)$, changes over minutes to hours, so within the analysis window h_l is a known constant (0.8585).
h_p	random	The static part (beam spread, A_0) is fixed by geometry, but building sway, thermal expansion and vibration make the beam centre jitter randomly at the receiver. The horizontal and vertical displacements are modelled as zero-mean Gaussian (σ_s), so the radial offset is Rayleigh and the collected fraction h_p is a random variable with PDF (7).
h_a	random	Temperature and pressure fluctuations along the path create random refractive-index eddies that constantly change. The log-amplitude is the sum of many small independent perturbations, hence Gaussian by the central-limit theorem, so h_a is log-normal with PDF (5).

Hence only h_p and h_a cause fading; h_l acts as a fixed scaling of the received power. Both random factors vary on millisecond time scales (coherence time of a few ms), which is very slow compared with the nanosecond symbol duration.

The three components represent independent multiplicative channel effects and therefore combine as $h = h_a h_p h_l$. The paper assumes fading is slow relative to the symbol rate and uses an infinite-interleaver assumption to obtain a memoryless channel model.

1.2 Turbulence PDF, eq. (5)

The PDF of h_a follows from the physics of weak turbulence in five steps.

Step 1 — Received field (Rytov model). In weak turbulence the optical field at the receiver is the free-space field U_0 multiplied by a random perturbation:

$$U = U_0\, e^{\chi + jS},$$

where χ is the *log-amplitude* fluctuation and S the phase fluctuation. Only χ affects the received power; S is discarded by the direct-detection (intensity) receiver.

Step 2 — χ is Gaussian. Along the 3 km path the beam crosses a very large number of small, independent turbulent eddies. Each eddy adds a small contribution to χ, so χ is a sum of many independent random terms. By the central-limit theorem,

$$\chi \sim \mathcal{N}(\mu_\chi, \sigma_X^2), \qquad f_\chi(\chi) = \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp\!\left(-\frac{(\chi-\mu_\chi)^2}{2\sigma_X^2}\right).$$

Step 3 — From field to irradiance. The photodetector responds to optical power, i.e. to $|U|^2$. Normalising by the turbulence-free power,

$$h_a = \frac{|U|^2}{|U_0|^2} = \left|e^{\chi + jS}\right|^2 = e^{2\chi}.$$

So $\ln h_a = 2\chi$: the logarithm of h_a is Gaussian, which is why h_a is called *log-normal*.

Step 4 — Fix the mean using $\mathbb{E}[h_a] = 1$. Turbulence only redistributes the received power in time; it does not add or remove power on average. Hence $\mathbb{E}[h_a] = 1$. For a Gaussian χ, the moment generating function gives

$$\mathbb{E}[h_a] = \mathbb{E}\big[e^{2\chi}\big] = e^{2\mu_\chi + \frac{1}{2}(2)^2\sigma_X^2} = e^{2\mu_\chi + 2\sigma_X^2}.$$

Setting this equal to 1:

$$2\mu_\chi + 2\sigma_X^2 = 0 \quad \Longrightarrow \quad \mu_\chi = -\sigma_X^2.$$

Therefore

$$\ln h_a = 2\chi \sim \mathcal{N}(2\mu_\chi,\ 4\sigma_X^2) = \mathcal{N}(-2\sigma_X^2,\ 4\sigma_X^2).$$

Step 5 — Change of variable to get $f_{H_a}(h_a)$. With $h_a = e^{2\chi}$, the inverse is $\chi = \frac{1}{2}\ln h_a$ and its derivative is $\dfrac{d\chi}{dh_a} = \dfrac{1}{2h_a}$. The transformation rule for PDFs gives

$$f_{H_a}(h_a) = f_\chi\Big(\tfrac{1}{2}\ln h_a\Big)\left|\frac{d\chi}{dh_a}\right| = \frac{1}{\sqrt{2\pi\sigma_X^2}}\exp\left(-\frac{\big(\frac{1}{2}\ln h_a + \sigma_X^2\big)^2}{2\sigma_X^2}\right)\frac{1}{2h_a}.$$

Multiplying numerator and denominator inside the exponent by 4, $\big(\frac{1}{2}\ln h_a + \sigma_X^2\big)^2/(2\sigma_X^2) = (\ln h_a + 2\sigma_X^2)^2/(8\sigma_X^2)$, so

$$\boxed{f_{H_a}(h_a) = \frac{1}{2h_a\sqrt{2\pi\sigma_X^2}}\exp\left(-\frac{(\ln h_a + 2\sigma_X^2)^2}{8\sigma_X^2}\right), \qquad h_a > 0,}$$

which is exactly eq. (5).

Link to the Rytov variance, eq. (3). The strength of the fluctuations is measured by the scintillation index $\sigma_I^2 = \mathbb{E}[h_a^2]/\mathbb{E}[h_a]^2 - 1 = e^{4\sigma_X^2} - 1$. For weak turbulence $4\sigma_X^2 \ll 1$, so $\sigma_I^2 \approx 4\sigma_X^2$. In weak turbulence the scintillation index equals the Rytov variance, $\sigma_I^2 \approx \sigma_R^2$, hence

$$\sigma_X^2 \approx \frac{\sigma_R^2}{4}, \qquad \sigma_R^2 = 1.23\, C_n^2 k^{7/6} L^{11/6} \text{ (eq. 4)}.$$

For Table I ($\lambda = 1550$ nm, $L = 3$ km), $\sigma_R^2 = 0.1, 0.5, 1$ correspond to $C_n^2 = 0.67, 3.4, 6.7 \times 10^{-15}\,\mathrm{m}^{-2/3}$, i.e. $\sigma_X^2 = 0.025, 0.125, 0.25$.

1.3 Pointing-error PDF, eqs. (6)–(7)

The PDF of h_p follows from the beam geometry and a Gaussian model of the jitter in six steps.

Step 1 — Collected power for a given offset. At the receiver plane the beam has a Gaussian intensity profile of radius w_z. If the beam centre is displaced by a radial distance r from the centre of the circular aperture $\mathcal{A}$ (radius a), the fraction of power collected is

$$h_p(r) = \int_{\mathcal{A}} \frac{2}{\pi w_z^2}\exp\left(-\frac{2\|\boldsymbol{\rho} - \boldsymbol{r}\|^2}{w_z^2}\right) d\boldsymbol{\rho}.$$

This integral has no closed form.

Step 2 — Gaussian approximation, eq. (6). Farid and Hranilovic [3] showed that, for a beam much wider than the aperture ($w_z \gg a$; here $w_z/a \approx 40$), $h_p(r)$ is itself very close to a Gaussian in r:

$$h_p(r) \approx A_0 \exp\left(-\frac{2r^2}{w_{z,eq}^2}\right), \qquad A_0 = \mathrm{erf}(v)^2, \quad v = \frac{\sqrt{\pi}\,a}{\sqrt{2}\,w_z}, \quad w_{z,eq}^2 = w_z^2\frac{\sqrt{\pi}\,\mathrm{erf}(v)}{2v\,e^{-v^2}}.$$

A_0 is the collected fraction with perfect alignment ($r = 0$) and $w_{z,eq}$ is the equivalent beam width. With Table I: $v = 0.0316$, $A_0 = 1.27 \times 10^{-3}$, $w_{z,eq} = 1.981$ m.

Step 3 — Statistics of the radial offset. The horizontal and vertical displacements are independent, zero-mean Gaussian with the same variance: $x, y \sim \mathcal{N}(0, \sigma_s^2)$. Then $R = \sqrt{x^2 + y^2}$ is Rayleigh distributed,

$$f_R(r) = \frac{r}{\sigma_s^2} \exp\left(-\frac{r^2}{2\sigma_s^2}\right), \qquad r \geq 0,$$

and R^2 is exponential with mean $2\sigma_s^2$, i.e. $\Pr\{R^2 > t\} = e^{-t/(2\sigma_s^2)}$.

Step 4 — Define the exponent V. Write eq. (6) as $h_p = A_0 e^{-V}$ with

$$V \coloneqq \frac{2R^2}{w_{z,eq}^2}.$$

Its tail probability is

$$\Pr\{V > v\} = \Pr\left\{R^2 > \frac{v\, w_{z,eq}^2}{2}\right\} = \exp\left(-\frac{v\, w_{z,eq}^2}{4\sigma_s^2}\right) = e^{-\gamma^2 v}, \qquad \gamma^2 = \frac{w_{z,eq}^2}{4\sigma_s^2},$$

so V is exponential with rate γ^2, where $\gamma = w_{z,eq}/(2\sigma_s)$ is eq. (8).

Step 5 — CDF of h_p. Since $h_p = A_0 e^{-V}$ decreases as V increases, for $0 \leq h_p \leq A_0$:

$$F_{H_p}(h_p) = \Pr\{A_0 e^{-V} \leq h_p\} = \Pr\left\{V \geq \ln \frac{A_0}{h_p}\right\} = \exp\left(-\gamma^2 \ln \frac{A_0}{h_p}\right) = \left(\frac{h_p}{A_0}\right)^{\gamma^2}.$$

Step 6 — Differentiate to get eq. (7).

$$\boxed{f_{H_p}(h_p) = \frac{d}{dh_p}\left(\frac{h_p}{A_0}\right)^{\gamma^2} = \frac{\gamma^2}{A_0^{\gamma^2}}\, h_p^{\gamma^2 - 1}, \qquad 0 \leq h_p \leq A_0.}$$

It integrates to $F_{H_p}(A_0) = 1$, as required.

Interpretation. γ compares the beam width with the jitter. For the three values of σ_s, $\gamma^2 = 68.1, 15.7, 8.0$. Large γ (beam much wider than the jitter) concentrates h_p near A_0; small γ spreads it towards zero. Figure 1 confirms (5) and (7) against samples drawn from the physical model (Gaussian log-amplitude; Gaussian x, y mapped through (6)).

1.4 Composite PDF (10) and a useful change of variable

The composite PDF follows by working with logarithms, where the product $h = h_l h_p h_a$ becomes a sum.

Step 1 — Product becomes a sum. Using $h_p = A_0 e^{-V}$,

$$\ln h = \ln h_l + \ln A_0 - V + \ln h_a \quad \Longrightarrow \quad u \coloneqq \ln \frac{h}{A_0 h_l} = \ln h_a - V,$$

with $\ln h_a \sim \mathcal{N}(-2\sigma_X^2, 4\sigma_X^2)$ (Sec. 1.2) and $V \sim \text{Exp}(\gamma^2)$ (Sec. 1.3), independent of each other.

Step 2 — Standardise the Gaussian part. Write $\ln h_a = -2\sigma_X^2 + 2\sigma_X W$ with $W \sim \mathcal{N}(0, 1)$, and define

$$z \coloneqq \frac{u + 2\sigma_X^2}{2\sigma_X} = W - E', \qquad E' \coloneqq \frac{V}{2\sigma_X}. \tag{2}$$

Scaling an exponential keeps it exponential: $\Pr\{E' > e\} = \Pr\{V > 2\sigma_X e\} = e^{-2\gamma^2 \sigma_X e}$, so

$$E' \sim \text{Exp}(\kappa), \qquad \boxed{\kappa = 2\gamma^2 \sigma_X = \gamma^2 \sigma_R} \quad (\text{since } \sigma_R = 2\sigma_X).$$

So in the variable z the channel is a standard Gaussian (turbulence) minus an independent exponential (pointing error), and the whole model depends on the single number κ.

Step 3 — PDF of z (convolution). For a difference of independent variables, condition on $E' = e$ (then $z = W - e$, so $W = z + e$):

$$f_Z(z) = \int_0^\infty f_{E'}(e)\, f_W(z+e)\, de = \int_0^\infty \kappa e^{-\kappa e}\, \phi(z+e)\, de, \qquad \phi(t) = \tfrac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

Completing the square in the exponent,

$$-\kappa e - \frac{(z+e)^2}{2} = -\frac{(z+e+\kappa)^2}{2} + \kappa z + \frac{\kappa^2}{2},$$

so

$$f_Z(z) = \kappa\, e^{\kappa z + \kappa^2/2} \int_0^\infty \phi(z+\kappa+e)\, de = \kappa\, e^{\kappa z + \kappa^2/2}\, Q(z+\kappa). \tag{3}$$

This is the exponentially-modified Gaussian (EMG) density.

Step 4 — Back to h: this is eq. (10). Since $h = A_0 h_l\, e^{2\sigma_X z - 2\sigma_X^2}$, $dz/dh = 1/(2\sigma_X h)$ and $f_H(h) = f_Z(z)/(2\sigma_X h) = \frac{\gamma^2}{h} e^{\kappa z + \kappa^2/2} Q(z+\kappa)$. To compare with eq. (10), check its three parts:

- $\dfrac{h^{\gamma^2-1}}{(A_0 h_l)^{\gamma^2}} = \dfrac{e^{\gamma^2 u}}{h}$, and $\gamma^2 u + 2\sigma_X^2\gamma^2(1+\gamma^2) = \gamma^2(2\sigma_X z - 2\sigma_X^2) + 2\sigma_X^2\gamma^2 + 2\sigma_X^2\gamma^4 = \kappa z + \dfrac{\kappa^2}{2}$;
- with $\mu = 2\sigma_X^2(1+2\gamma^2)$: $\dfrac{u+\mu}{\sqrt{8\sigma_X^2}} = \dfrac{2\sigma_X z + 2\sigma_X \kappa}{2\sqrt{2}\,\sigma_X} = \dfrac{z+\kappa}{\sqrt{2}}$;
- $\operatorname{erfc}(\frac{z+\kappa}{\sqrt{2}}) = 2Q(z+\kappa)$.

Putting these into eq. (10) gives $\frac{\gamma^2}{2h} \cdot e^{\kappa z + \kappa^2/2} \cdot 2Q(z+\kappa)$, identical to the line above. So eq. (10) and (3) are the same function.

Step 5 — CDF in closed form.

$$F_Z(z) = \Pr\{W \le z + E'\} = \int_0^\infty \kappa e^{-\kappa e}\, \Phi(z+e)\, de.$$

Integrating by parts ($\Phi' = \phi$):

$$F_Z(z) = \Big[-e^{-\kappa e}\Phi(z+e)\Big]_0^\infty + \int_0^\infty e^{-\kappa e}\phi(z+e)\, de = \Phi(z) + e^{\kappa z + \kappa^2/2} Q(z+\kappa). \tag{4}$$

This gives the outage probability of Task 4 directly.

Meaning of κ. $1/\kappa = \mathbb{E}[E']$ is the mean pointing fade measured in units of the turbulence standard deviation. Large κ: pointing loss is small compared with the turbulence spread and the channel is nearly log-normal. Small κ: pointing loss dominates. Table 1 lists κ and the other derived parameters.

1.5 Why (10) is numerically unstable for small σ_s

Step 1 — Size of each factor. Written as in the paper, eq. (10) is a product of four factors. Evaluate them at the green point ($\sigma_s = 0.12$ m, $\sigma_R^2 = 1$, so $\gamma^2 = 68.1$, $\sigma_X = 0.5$, $\kappa = 68.1$) and at a typical gain $h = A_0 h_l = 1.09 \times 10^{-3}$:

$$\underbrace{e^{2\sigma_X^2\gamma^2(1+\gamma^2)}}_{e^{2353}} \cdot \underbrace{(A_0 h_l)^{-\gamma^2}}_{e^{464}} \cdot \underbrace{h^{\gamma^2-1}}_{e^{-458}} \cdot \underbrace{\operatorname{erfc}\Big(\tfrac{\ln(h/A_0 h_l)+\mu}{\sqrt{8\sigma_X^2}}\Big)}_{e^{-2358}} \cdot \frac{\gamma^2}{2} \;\approx\; e^{1.8} \cdot 34 \approx 2 \times 10^2.$$

The product is an ordinary number, but no single factor can be stored.

Table 1: Derived parameters at the nine operating points ($\theta_s = \sigma_s/L$).

σ_s [m]	θ_s [mrad]	σ_R^2	γ^2	μ	$\hat{h}$	$\kappa = \gamma^2\sigma_R$	$2\sigma_X^2\gamma^2(1+\gamma^2)$
0.12	0.040	0.1	68.1	6.86	1.1×10^{-6}	21.5	235
0.12	0.040	0.5	68.1	34.3	1.4×10^{-18}	48.2	1177
0.12	0.040	1.0	68.1	68.6	1.8×10^{-33}	**68.1**	**2353**
0.25	0.083	0.1	15.7	1.62	2.2×10^{-4}	4.96	13.1
0.25	0.083	0.5	15.7	8.10	3.3×10^{-7}	**11.1**	65.5
0.25	0.083	1.0	15.7	16.2	1.0×10^{-10}	15.7	131
0.35	0.117	0.1	8.01	0.85	4.7×10^{-4}	**2.53**	3.6
0.35	0.117	0.5	8.01	4.25	1.6×10^{-5}	5.66	18.0
0.35	0.117	1.0	8.01	8.51	2.2×10^{-7}	8.01	36.1

Step 2 — Limits of double precision. A double can only represent numbers between about e^{-745} and $e^{709.8}$. So `exp(2353)` returns `inf`, `erfc`(48.5) returns 0, and the product is `inf`$\cdot 0 =$ `NaN`.

Step 3 — Why it gets worse for small σ_s. The largest exponent is $2\sigma_X^2\gamma^2(1+\gamma^2) \approx 2\sigma_X^2\gamma^4 = \kappa^2/2$, with $\gamma^2 = w_{z,eq}^2/(4\sigma_s^2)$. Halving σ_s multiplies γ^4 by 16, and stronger turbulence (larger σ_X^2) increases it further. The erfc argument $\approx \kappa/\sqrt{2}$ grows at the same rate, so its value underflows at the same time.

Step 4 — Where it fails. Overflow starts when $2\sigma_X^2\gamma^2(1+\gamma^2) = 709.8$ (dashed line in Fig. 4). The red region, found by evaluating (10) literally on a grid, is slightly larger because `erfc` underflows (argument > 26.5) before `exp` overflows. At $\sigma_R^2 = 1$ the formula fails for $\sigma_s \lesssim 0.17$ m.

Step 5 — The fix. The huge and tiny factors cancel analytically, so combine the exponents before evaluating anything. Using (3) in the log domain,

$$\ln f_Z(z) = \ln\kappa + \kappa z + \frac{\kappa^2}{2} + \ln Q(z+\kappa),$$

where $\ln Q(\cdot)$ is computed directly with a log-Q routine (`scipy.special.log_ndtr`), which never underflows. The $+\kappa^2/2$ and $\ln Q \approx -\kappa^2/2$ cancel as moderate numbers, and only the final result is exponentiated. This is exact and stable for every (σ_s, σ_R^2).

1.6 The approximation (13)–(14)

Step 1 — The erfc approximation, eq. (13). For large x, $\mathrm{erfc}(x)$ has the asymptotic series

$$\mathrm{erfc}(x) = \frac{e^{-x^2}}{x\sqrt{\pi}}\left(1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \dots\right).$$

Eq. (13) keeps only the first term, $\mathrm{erfc}(x) \approx e^{-x^2}/(x\sqrt{\pi})$. It is an upper bound for $x > 0$ with relative error $\approx 1/(2x^2)$: very accurate for large x, but it diverges as $x \to 0$. In terms of $Q(y) = \frac{1}{2}\mathrm{erfc}(y/\sqrt{2})$ this reads $Q(y) \approx \phi(y)/y$.

Step 2 — Insert into the composite PDF. In (3) the argument is $y = z + \kappa$, so

$$f_Z(z) \approx \kappa\, e^{\kappa z + \kappa^2/2}\, \frac{\phi(z+\kappa)}{z+\kappa}.$$

Step 3 — Merge the exponents. Since $\frac{(z+\kappa)^2}{2} = \frac{z^2}{2} + \kappa z + \frac{\kappa^2}{2}$, we have $e^{\kappa z + \kappa^2/2}\phi(z+\kappa) = \phi(z)$. All γ-dependent exponentials cancel:

$$f_Z(z) \approx \frac{\kappa\,\phi(z)}{z+\kappa} = \frac{\phi(z)}{1+z/\kappa}, \qquad z > -\kappa. \tag{5}$$

Transforming back with $f_H = f_Z/(2\sigma_X h)$ gives exactly eq. (14) of the paper.

Step 4 — Interpretation. (5) is the log-normal turbulence PDF $\phi(z)$ (shifted by the deterministic loss $A_0 h_l$) times a correction $1/(1+z/\kappa)$ for the pointing error. For $\kappa \to \infty$ the correction disappears and the approximation becomes exact. Written this way it contains no large exponentials and cannot overflow. (Coded term by term as printed, (14) still contains $e^{2\sigma_X^2\gamma^2(1+\gamma^2)}$ and overflows like (10); the stability comes from merging the exponents.)

Step 5 — Why the support is cut at $\hat{h}$. The erfc argument is $x = (z+\kappa)/\sqrt{2}$. It is zero when $z = -\kappa$, i.e. $u = -\mu$, i.e.

$$h = \hat{h} = A_0 h_l\, e^{-\mu}.$$

For $h < \hat{h}$, $x < 0$ and $e^{-x^2}/(x\sqrt{\pi}) < 0$, while the true erfc$(x)$ lies between 1 and 2. The approximate PDF would be negative there, so it is set to zero. $\hat{h}$ is found by setting the erfc argument to zero, as in the paper.

Step 6 — Is the area still 1? No. Three effects change it (quantified in Task 3):

1. *Lost mass.* The exact probability below $\hat{h}$ is discarded. From (4), $F_Z(-\kappa) = \Phi(-\kappa) + e^{-\kappa^2/2}Q(0) = Q(\kappa) + \frac{1}{2}e^{-\kappa^2/2}$: 2.6 % for the pink point, 3×10^{-6} for $(0.25, 0.1)$, $< 10^{-7}$ for the others.

2. *Over-estimate above $\hat{h}$.* Because (13) is an upper bound, the approximate PDF is too large. Expanding $\frac{1}{1+z/\kappa} = 1 - \frac{z}{\kappa} + \frac{z^2}{\kappa^2} - \frac{z^3}{\kappa^3} + \frac{z^4}{\kappa^4} - \ldots$ and integrating against $\phi(z)$ (odd moments vanish, $\mathbb{E}[W^2] = 1$, $\mathbb{E}[W^4] = 3$):

$$\text{area} \approx 1 + \frac{1}{\kappa^2} + \frac{3}{\kappa^4} + \ldots$$

3. *Pole at $\hat{h}$.* Near $z = -\kappa$ the integrand behaves like $\kappa\phi(\kappa)/(z+\kappa)$. Integrating from $z = -\kappa+\delta$,

$$\int_{-\kappa+\delta} \frac{\kappa\phi(\kappa)}{z+\kappa}\,dz \;\sim\; \kappa\phi(\kappa)\ln\frac{1}{\delta} \;\to\; \infty \quad (\delta \to 0).$$

   So, strictly, the area under (14) is infinite. The weight $\kappa\phi(\kappa)$ is 4×10^{-2} at the pink point and at most 10^{-5} elsewhere, which is why the effect is invisible in the paper except as the "wrong slope" it reports for high pointing errors.

In practice the area is slightly above 1, and the excess shrinks like $1/\kappa^2$.

2 Task 2 — Reproducing Fig. 2 (right)

2.1 Implementation

Python (NumPy/SciPy), described in detail in the separate code document.

- **Monte Carlo** (`montecarlo.py`): i.i.d. bits → OOK $\{0, 2P\}$; for every bit a fresh $h = h_l h_p h_a$ with $h_a = e^{2\chi}$, $\chi \sim \mathcal{N}(-\sigma_X^2, \sigma_X^2)$ and $h_p = A_0 e^{-2(x^2+y^2)/w_{z,eq}^2}$, $x, y \sim \mathcal{N}(0, \sigma_s^2)$ (the infinite interleaver makes fades independent from symbol to symbol, so Π, Π^{-1} need not be simulated); $y = \eta h x + n$; ML hard decision with perfect CSI, $\hat{b} = \mathbb{1}\{y > \eta h P\}$. 4×10^6 bits per power, seed 2024.

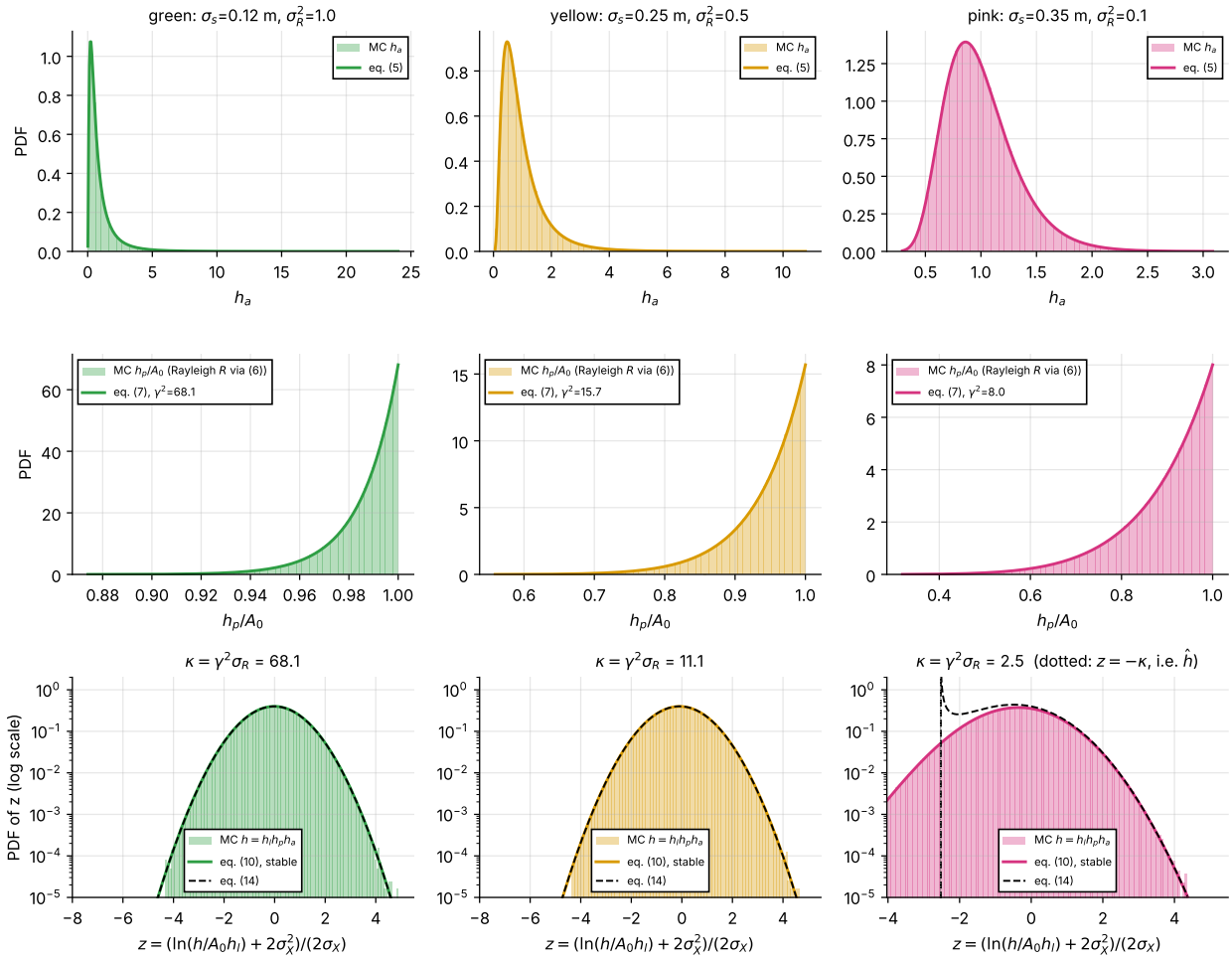


Figure 1: Model check, 2×10^6 samples per panel drawn from the *physical* model. Rows: h_a vs (5); h_p/A_0 vs (7); composite z vs stable (10) and (14). The dashed (14) is indistinguishable from (10) for $\kappa = 68$ and 11; for $\kappa = 2.5$ it overshoots and has the pole at $z = -\kappa$ (dotted).

- **Exact BER (16)** two ways: (a) literally, `scipy.integrate.quad` over h with (10) as printed; (b) stably, Simpson's rule over z with (3) in the log domain.
- **Approximation (17)**: $\int_{z>-\kappa} P_b(h(z))\,\kappa\phi(z)/(z+\kappa)\,dz$, computed with $t = \ln(z+\kappa)$ (which turns the pole into a smooth, infinite range) and a lower limit $z = -\kappa + \delta$, $\delta = 10^{-6}$. Eq. (18) likewise.
- Δ = (power at which the approximation reaches 3.84×10^{-3}) − (same for exact (16)), by log-linear interpolation on a 0.05–0.1 dB grid.

2.2 Results

Table 2: Power at the HD-FEC limit and gaps Δ [dB]. MC crossing interpolated between 1 dB-spaced points (interpolation error $\sim$ 0.01–0.02 dB).

Point (σ_s, σ_R^2)	κ	P_{16} [dBm]	MC − (16)	Δ_{17}	Δ_{17} paper	Δ_{18}	Δ_{18} paper
green (0.12, 1.0)	68.1	5.66	−0.01	0.0004	≈ 0	0.75	≈ 0.7
yellow (0.25, 0.5)	11.1	2.27	−0.02	0.018	0.02	0.44	≈ 0.4
pink (0.35, 0.1)	2.53	−1.07	−0.01	1.30†	0.18	1.37	$\approx$ (17)

† lower limit $\delta = 10^{-6}$; 0.21 dB with $\delta = 0.1$ (Sec. 2.3).

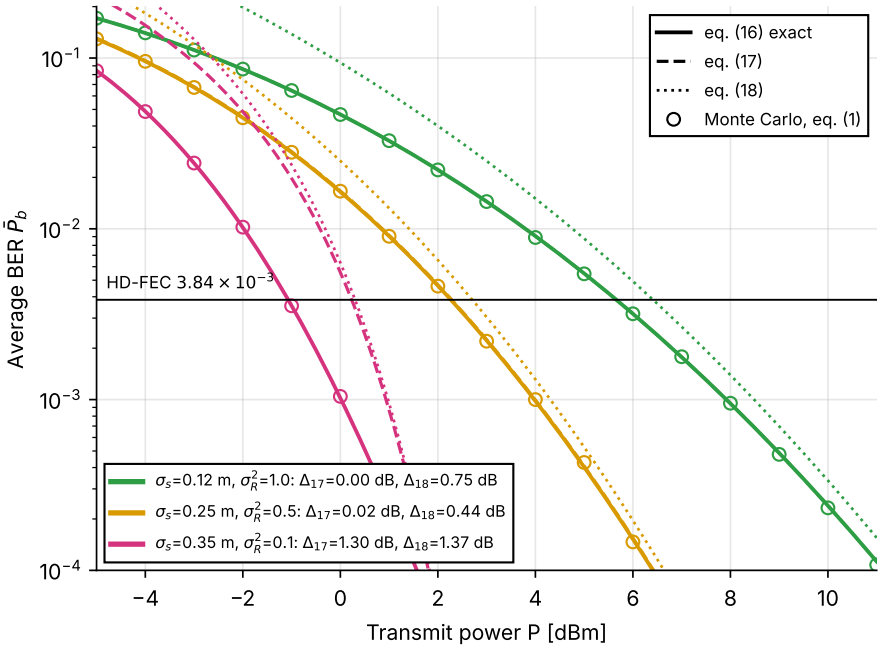


Figure 2: Reproduction of Fig. 2 (right). Solid: exact (16), stable evaluation; dashed: (17); dotted: (18); circles: bit-level Monte Carlo of Fig. 1 (4×10^6 bits per point). Pink: Δ_{17} shown for $\delta = 10^{-6}$, see Sec. 2.3.

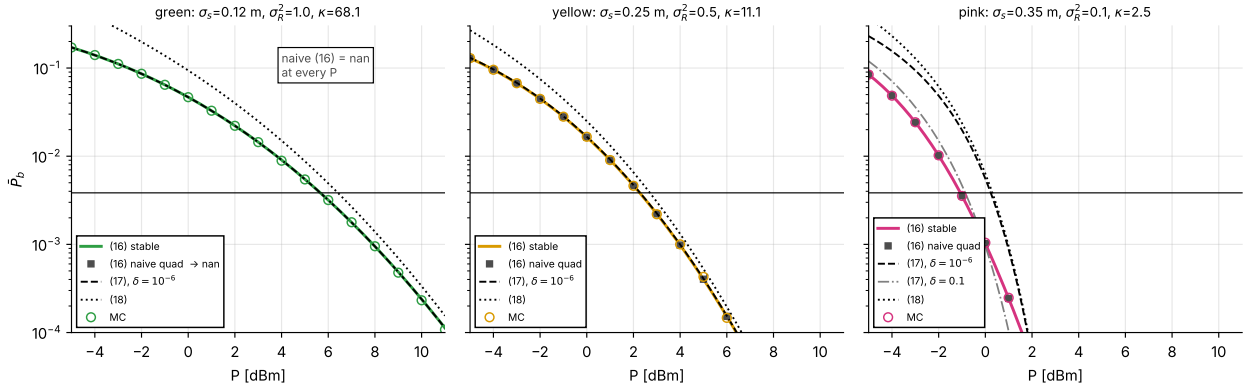


Figure 3: Same curves per operating point, with the literal ("naive") evaluation of (16) (squares). Green: the naive integral returns NaN at every power; the stable (16) passes through the Monte Carlo points. Pink: (17) for two lower limits δ.

- **Monte Carlo vs exact (16).** At 41 simulated powers the MC BER lies within 2.7 standard errors of the stable (16) at 40; the remaining one (green, 4 dBm, 3.6σ) was re-simulated with three other seeds and 8×10^6 bits and fell within 1σ. This validates the stable (16) also at the green point, where the paper could only show Monte Carlo.
- **Green and yellow.** $\Delta_{17} = 0.0004$ and 0.018 dB, $\Delta_{18} = 0.75$ and 0.44 dB, as in the paper (0, 0.02, ≈ 0.7, ≈ 0.4).
- **(16) at the green point.** Coded literally, (16) returns NaN at every power (Fig. 3, left) for the reason in Sec. 1.5 ($e^{2353} \cdot 0$). The paper's red region is reproduced in Fig. 4: (10) breaks for $\sigma_s \lesssim 0.17$ m at $\sigma_R^2 = 1$ and for $\sigma_s \lesssim 0.11$ m at $\sigma_R^2 = 0.1$. Evaluated in the log domain, (16) is finite everywhere and needs no approximation at all.

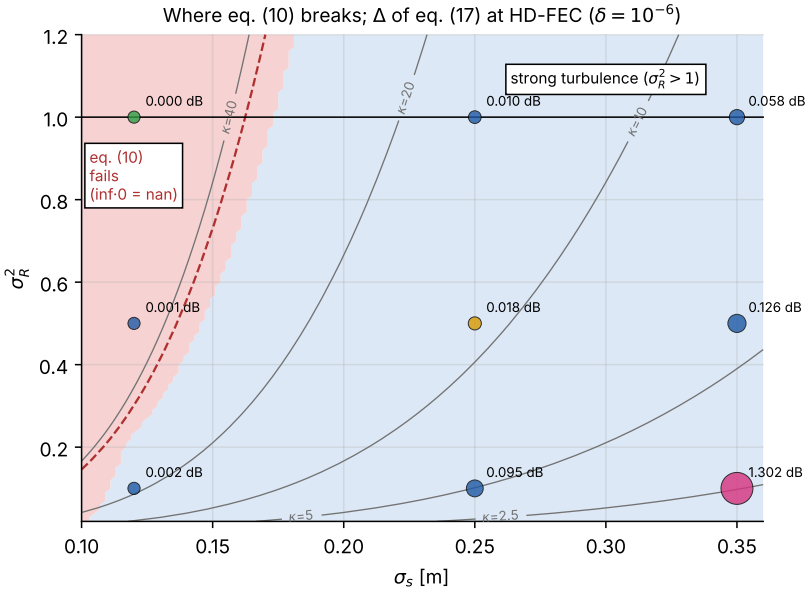


Figure 4: Fig. 2 (left) reproduced. Red: where (10) coded literally gives NaN or loses the density; dashed: overflow boundary $2\sigma_X^2\gamma^2(1+\gamma^2) = 709.8$; grey: contours of $\kappa = \gamma^2\sigma_R$. Labels: Δ_{17} at the nine points ($\delta = 10^{-6}$); marker size $\propto \Delta_{17}$.

2.3 The pink point: Δ_{17} is not unique

For $\kappa = 2.5$ the result of (17) depends on how close to $\hat{h}$ the integral starts, because of the logarithmic divergence of Sec. 1.6: lower limits $\delta = 10^{-1}, 10^{-6}, 10^{-10}$ give $\Delta_{17} = 0.21$, 1.30, 1.60 dB. The paper's 0.18 dB corresponds to stopping about $\delta \approx 0.1$ from $\hat{h}$, which a coarse grid or a quadrature routine that gives up near a singular endpoint would do. The divergent term adds roughly $\kappa\phi(\kappa)\, P_b(\hat{h})\ln(1/\delta)$; since $P_b(\hat{h})$ decays more slowly with P than the true BER, it also tilts the curve, which is the incorrect slope the paper notes for high pointing errors. Eq. (18) inherits the same pole.

3 Task 3 — Checking the approximation

Assumption: the nine points of Fig. 2 (left) are the grid $\sigma_s \in \{0.12, 0.25, 0.35\}$ m $\times\ \sigma_R^2 \in \{0.1, 0.5, 1\}$, consistent with the three highlighted points and with the Δ labels.

Table 3: Area under (14) integrated over $z > -\kappa + \delta$, divergence rate $\kappa\phi(\kappa)$, exact mass discarded below $\hat{h}$, and BER gaps at HD-FEC [dB]. "Remedy": bounded Q approximation, Sec. 3.

σ_s	σ_R^2	κ	area (14), $\delta =$ 10^{-2}	10^{-6}	10^{-10}	$\kappa\phi(\kappa)$	lost mass	$1+\frac{1}{\kappa^2}+\frac{3}{\kappa^4}$	Δ_{17}	paper	Δ_{18}	remedy $\lvert\Delta\rvert$
0.12	0.1	21.5	1.0022	1.0022	1.0022	2×10^{-100}	10^{-101}	1.0022	0.002	0.005	0.14	0.0001
0.12	0.5	48.2	1.0004	1.0004	1.0004	≈ 0	≈ 0	1.0004	0.001	0.003	0.42	0.0000
0.12	1.0	68.1	1.0002	1.0002	1.0002	≈ 0	≈ 0	1.0002	0.000	0	0.75	0.0000
0.25	0.1	4.96	1.0471	1.0472	1.0473	9×10^{-6}	3×10^{-6}	1.0456	0.095	0.11	0.24	0.0020
0.25	0.5	11.1	1.0083	1.0083	1.0083	8×10^{-27}	10^{-27}	1.0083	0.018	0.02	0.44	0.0010
0.25	1.0	15.7	1.0041	1.0041	1.0041	2×10^{-53}	2×10^{-54}	1.0041	0.010	0.01	0.77	0.0006
0.35	0.1	2.53	**1.374**	**1.752**	**2.129**	**0.041**	**0.026**	(1.23)	**1.30**	0.18	1.37	0.0003
0.35	0.5	5.66	1.0347	1.0347	1.0347	2×10^{-7}	6×10^{-8}	1.0341	0.126	0.13	0.57	0.0035
0.35	1.0	8.01	1.0164	1.0164	1.0164	4×10^{-14}	7×10^{-15}	1.0163	0.058	0.06	0.83	0.0029

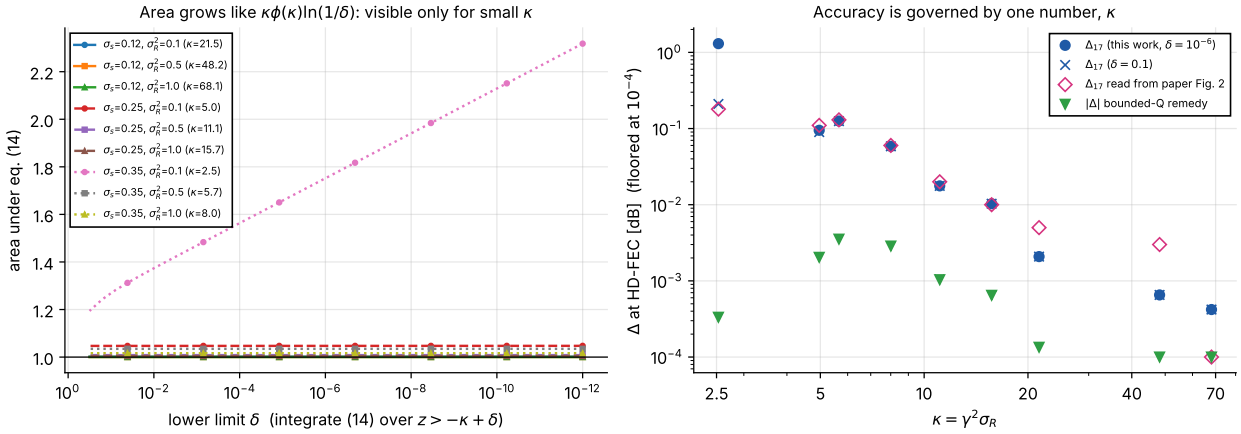


Figure 5: Left: area under (14) as the lower limit approaches $\hat{h}$; only the pink point ($\kappa = 2.5$) shows the logarithmic growth. Right: Δ against κ; this work vs. values read from the paper, and the remedy.

Relation between area and accuracy. The area excess and Δ_{17} move together (Table 3, Fig. 5): excess $0.0002 \to 0.047$ as κ goes $68 \to 5$, and Δ_{17} goes $0.0004 \to 0.095$ dB. This is expected: the extra area lies at the low-h side ($z < 0$, where $1/(1+z/\kappa) > 1$), i.e. in the deep fades that dominate the average BER, so an approximate PDF that is too heavy there over-estimates the BER and pushes the HD-FEC crossing to higher power ($\Delta > 0$ at all points). For eight points the area is fully explained by the series $1+1/\kappa^2+3/\kappa^4$ and is independent of δ (the parenthesised pink entry is the asymptotic series, which is no longer meaningful at $\kappa = 2.5$); at the pink point it is dominated by the pole, the lost mass is 2.6 %, and Δ_{17} is not well defined.

Why it improves with more turbulence and smaller pointing errors. Everything depends on $\kappa = \gamma^2\sigma_R$, which increases with σ_R (turbulence) and with $\gamma^2 = w_{z,eq}^2/(4\sigma_s^2)$ (smaller pointing error). Three views of the same fact:

1. The erfc argument over the bulk of the distribution ($z = O(1)$) is $x = (z+\kappa)/\sqrt{2} \approx \kappa/\sqrt{2}$. The relative error of (13) is $\approx 1/(2x^2) \approx 1/\kappa^2$: 2×10^{-4} for green, 8×10^{-3} for yellow, 0.16 for pink.

2. The pole sits at $z = -\kappa$, i.e. κ standard deviations below the mean of the log-gain; its weight is $\kappa\phi(\kappa) = 4 \times 10^{-2}$, 8×10^{-27}, ≈ 0 for pink, yellow, green.

3. Physically, $1/\kappa$ is the mean pointing fade measured in units of the turbulence spread. If the pointing fade is small compared with the log-normal spread, the composite is nearly log-normal and (14) $= \phi(z)/(1+z/\kappa) \to \phi(z)$ becomes exact. Increasing turbulence does not make the channel better; it makes the pointing contribution relatively smaller, which is the part (13) approximates badly.

With my numbers: going from $\sigma_R^2 = 0.1$ to 1 at $\sigma_s = 0.35$ m raises κ from 2.5 to 8.0 and lowers Δ_{17} from 1.30 (ill-defined) to 0.058 dB; going from $\sigma_s = 0.35$ to 0.12 m at $\sigma_R^2 = 0.5$ raises κ from 5.7 to 48 and lowers Δ_{17} from 0.126 to 0.0007 dB.

Remedy check. The trouble comes entirely from $\operatorname{erfc}(x) \approx e^{-x^2}/(x\sqrt{\pi})$ near $x = 0$. Replacing it by the bounded Börjesson–Sundberg form $Q(y) \approx \phi(y)/((1-a)y + a\sqrt{y^2+b})$, $a = 0.339$, $b = 5.51$ (eq. 13 is $a = 0$), and using $Q(y) = 1 - Q(-y)$ for $y < 0$, gives a PDF $\kappa\phi(z)/((1-a)(z+\kappa) + a\sqrt{(z+\kappa)^2+b})$ that is finite at $\hat{h}$ and defined for all h. Its area is within 0.2 % of 1 and $|\Delta| < 0.004$ dB at all nine points, including pink (last column of Table 3), so the slope problem disappears. The cost is one square root; the BER integral still has no closed form.

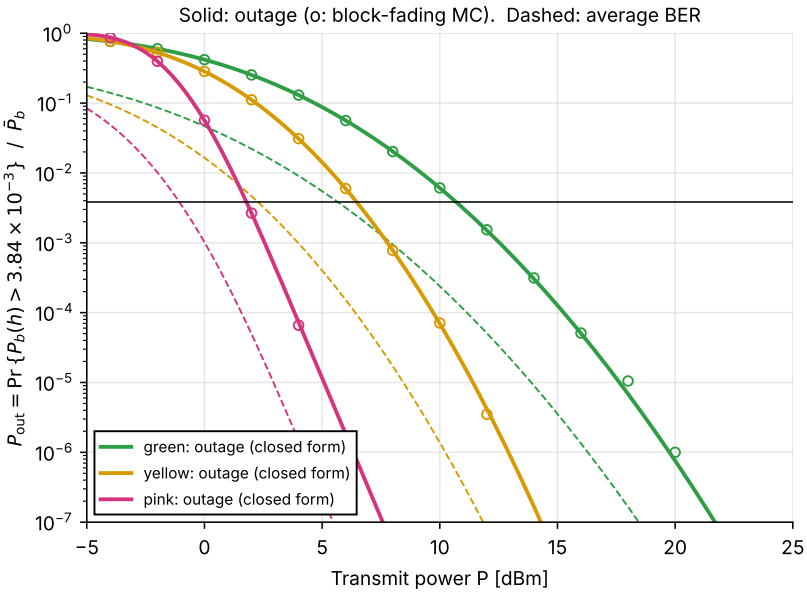


Figure 6: Outage probability (solid, closed form; circles, block-fading Monte Carlo with 2×10^6 blocks) vs. average BER (dashed) for the three operating points.

4 Task 4 (stretch, option b) — Outage instead of average BER

Time scales. Coherence time T_c = 1–5 ms (paper: milliseconds). I take a symbol rate R_s = 10 GBd (10 Gb/s OOK). One fade then lasts $R_s T_c = 10^7$–5×10^7 symbols. Hard-decision FECs at rate ≈ 0.94 (staircase/product codes) have blocks of 10^4–10^6 bits, so *every codeword sees a single value of h*. An interleaver that makes symbols see independent fades would have to span many coherence times: N independent fades per codeword need NR_sT_c symbols of memory and NT_c of latency (for $N = 100$: 10^9 symbols and 100–500 ms). The infinite-interleaver assumption is therefore an idealisation.

Outage — derivation. Without such interleaving, the relevant event is that a block's conditional BER exceeds the HD-FEC limit.

1. *Conditional BER.* For a fixed gain h, eq. (15) gives $P_b(h) = \frac{1}{2}\operatorname{erfc}(\eta P h/(\sqrt{2}\sigma_n))$, which decreases monotonically with h.
2. *Threshold gain.* Hence $P_b(h) > 3.84 \times 10^{-3}$ is the same event as $h < h_{\text{th}}$, where $P_b(h_{\text{th}}) = 3.84 \times 10^{-3}$:
$$h_{\text{th}} = \frac{\sqrt{2}\,\sigma_n\ \operatorname{erfc}^{-1}(2 \cdot 3.84 \times 10^{-3})}{\eta P}.$$
3. *Map to z.* $z_{\text{th}} = (\ln \frac{h_{\text{th}}}{A_0 h_l} + 2\sigma_X^2)/(2\sigma_X)$.
4. *Use the CDF* (4).

$$P_{\text{out}}(P) = \Pr\{h < h_{\text{th}}\} = F_Z(z_{\text{th}}) = \Phi(z_{\text{th}}) + e^{\kappa z_{\text{th}} + \kappa^2/2} Q(z_{\text{th}} + \kappa), \tag{6}$$

This is exact and in closed form; Fig. 6 checks it against block-fading Monte Carlo.

Which quantity does a coded system care about? The outage probability (equivalently, the codeword/frame error rate). The decoder succeeds or fails per codeword according to the error rate *inside* that codeword; the average BER mixes good blocks (BER $\ll 10^{-3}$) with rare deep fades (BER near 0.5) into one number that no single codeword experiences. The average BER is the right pre-FEC metric only if each codeword is spread over enough independent fades for the law of

Table 4: Transmit power [dBm] needed by each design criterion.

Point	$\bar{P}_b = 3.84 \times 10^{-3}$	$P_{\text{out}} = 10^{-2}$	$P_{\text{out}} = 10^{-3}$	$P_{\text{out}} = 10^{-6}$	penalty @10^{-3}	penalty @10^{-6}
green	5.66	9.22	12.53	19.76	6.9	14.1
yellow	2.27	5.42	7.78	12.92	5.5	10.7
pink	−1.07	1.20	2.56	6.33	3.6	7.4

large numbers to apply. At the HD-FEC power the average-BER design leaves the link in outage 7 % (green), 10 % (yellow) and 19 % (pink) of the time; a 10^{-3} outage target costs 3.6–6.9 dB more power, and a 10^{-6} target 7–14 dB. The penalty grows with the spread of the fading: it is largest at green ($\sigma_R^2 = 1$, standard deviation of $\ln h_a$ equal to 1) and smallest at pink ($\sigma_R^2 = 0.1$, 0.32).

5 Task 5 — What next: how deep must the interleaver be?

Proposal. Replace the infinite-interleaver assumption by an interleaver of finite depth N (each codeword spread over N independent fading blocks) and derive, for this channel, the transmit power needed for a target codeword-failure probability as a function of N, i.e. of latency NT_c and memory NR_sT_c. Then use the result to co-design the FEC, the interleaver and the power budget, and to see where receiver-side CSI should be used to erase deeply faded blocks rather than to average over them.

Preliminary result. With the HD-FEC criterion applied to the codeword-average pre-FEC BER, $\Pr\{\frac{1}{N}\sum_{i=1}^{N} P_b(h_i) > 3.84 \times 10^{-3}\} = 10^{-3}$, Monte Carlo gives Fig. 7:

- $N = 1$ reproduces the closed-form outage (12.5, 7.8, 2.6 dBm for green, yellow, pink).
- Small N does *not* help and can hurt: for green the required power rises from 12.5 dBm ($N = 1$) to 13.4 dBm for $N = 4$–16. A block in a deep fade has BER near $\frac{1}{2}$; as long as $\frac{1}{2N} > 3.84 \times 10^{-3}$, i.e. $N \lesssim 130$, that single block fails the whole codeword, and spreading the codeword over more blocks only gives it more chances to hit one.
- The infinite-interleaver result is approached slowly: at $N = 1024$ the gap is still 1.1 dB (green, 6.8 vs 5.7 dBm), 0.65 dB (yellow) and 0.26 dB (pink). With $T_c = 1$ ms, $N = 1024$ means about one second of interleaving latency and 10^{10} symbols of memory at 10 GBd.

Why it matters.

- **It is the number a system designer needs.** The paper's model exists to design modulation and coding for real FSO, satellite and hybrid links. Those links have millisecond fades, gigabaud rates and latency budgets of tens of milliseconds, i.e. they sit in the finite-N regime, where the average-BER design is several dB optimistic (Task 4) and the gain from interleaving behaves non-intuitively.
- **The model makes it tractable.** In the z variable the channel is an exponentially-modified Gaussian with one shape parameter κ and a closed-form CDF. The $N = 1$ case is already exact (4); for $N > 1$ the codeword error rate is a tail probability of a sum of i.i.d. terms, for which Chernoff/saddle-point bounds with the EMG moment-generating function $\mathbb{E}[e^{sZ}] = e^{s^2/2}\kappa/(\kappa + s)$ give closed-form or one-line numerical answers. A (κ, N) design chart would replace Monte Carlo.
- **It points to better receivers.** The non-monotone curve says that a single deep fade (BER $\approx \frac{1}{2}$) dominates the failure probability until $N \gtrsim \frac{1}{2}/3.84 \times 10^{-3} \approx 130$. Since the receiver knows h, it could declare such blocks erased and let an outer erasure code recover them; the analysis above gives the right yardstick to quantify that gain, and extends naturally to 4-PAM

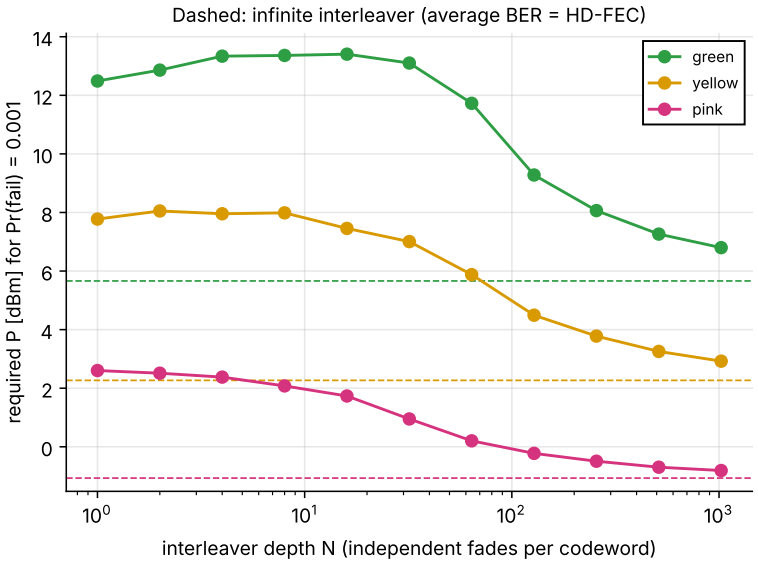


Figure 7: Power needed so that at most 10^{-3} of codewords exceed the HD-FEC limit, vs. the number N of independent fades per codeword. $N = 1$ equals the closed-form outage; dashed lines: infinite interleaver (average BER).

and to gamma-gamma turbulence (the EMG structure holds for any turbulence law whose logarithm has a known density convolved with the exponential pointing term).

Concrete next steps. (1) Chernoff/saddle-point approximation of the finite-N failure probability with the EMG MGF, validated against Fig. 7; (2) a realistic block-fading time series (correlated Gaussian jitter and turbulence with T_c of a few ms) instead of i.i.d. blocks; (3) replace the threshold criterion by the actual decoding behaviour of a staircase code, with and without CSI-based erasures; (4) repeat for a LEO downlink geometry with the group's satellite parameters.

Summary

- In $z = (\ln \frac{h}{A_0 h_l} + 2\sigma_X^2)/(2\sigma_X)$ the channel is $z = W - E'$, $E' \sim \mathrm{Exp}(\kappa)$, $\kappa = \gamma^2\sigma_R$; (10) is $\kappa e^{\kappa z + \kappa^2/2} Q(z + \kappa)$ and (14) is $\phi(z)/(1 + z/\kappa)$.
- (10) fails numerically because its factors are $e^{\pm\kappa^2/2}$-sized; evaluated in the log domain it is exact and stable, and the exact BER (16) agrees with Monte Carlo at all three points, including green.
- Fig. 2 is reproduced; Δ_{17} matches the paper within 0.015 dB at eight points. At the ninth ($\kappa = 2.5$), (14) has a non-integrable pole at $\hat{h}$, its area and (17) diverge logarithmically and Δ_{17} depends on the integration limit. A bounded Q approximation gives $|\Delta| < 0.004$ dB everywhere.
- Accuracy improves with turbulence and with smaller pointing error because both raise κ; the area excess is $\approx 1/\kappa^2$.
- Coded links without very deep interleaving are limited by outage, which is closed form here and needs 3.6–6.9 dB more power than the average-BER design (target 10^{-3}). Proposed follow-up: a finite-depth interleaver analysis of this channel.

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